

Introduction to Biotechnology and Pharmaceutical Science

Science

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Introduction to Biotechnology and Pharmaceutical Science (A07549)

Short Title: Intro to Biotech & Pharma Sci
Faculty Science and Computing
Department Science
Cluster Pharmaceutical Science
Author CLENNON
Credits 5

Level: Introductory

Description of Module / Aims

This module will provide the students with an appreciation of the pharmaceutical and biotechnology industries. Students will be introduced to the wide range of products produced by these industries. An overview of the processes and quality aspects of production will also be included.

Programmes

BIPH-0001	BSc (Common Entry) (WD_SSCIE_D)	1 / 2 / E
BIPH-0001	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 2 / E
BIPH-0001	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 2 / E
BIPH-0001	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	1 / 2 / M
BIPH-0001	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	1 / 2 / E
BIPH-0001	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 2 / E
BIPH-0001	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	1 / 2 / M
BIPH-0001	BSc in Pharmaceutical Science (WD_SPHSC_D)	1 / 2 / M

Indicative Content

- Role of regulatory authorities: USA FDA, HPRA, UK MHRA
- Introduction to pharmaceuticals: terminology and general requirements
- Classification of pharmaceuticals: topical, solid dose, parenterals and aerosols
- Chemical structures and synthesis of pharmaceutical products
- Overview of API processing and formulation development
- Historical and modern drug development
- Explanation of Good Manufacturing Practice (GMP) and its role in pharmaceutical production
- Roles, responsibilities and career prospects within the pharmaceutical industry
- Biotechnology in food, pharmaceuticals, agriculture and the environment
- Overview of a typical biotechnology process: inputs, process & outputs
- Brief introduction to the principles of genetic engineering of microbes, plants & animals and its potential
- Fundamentals of biologics and biopharmaceuticals

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the roles and structure within the pharmaceutical industry.
2. Describe a typical biotechnological and pharmaceutical process.
3. Identify the impact of regulation and GMP on pharmaceutical production.
4. Describe the evolution and discovery of pharmaceutical products.
5. Communicate scientific material to peers.
6. Illustrate effective teamwork in conjunction with colleagues.

Learning and Teaching Methods

- Lectures.
- Groupwork to produce presentations.
- Class discussions.
- Blended learning.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Presentation</i>	25%	5,6
<i>In-Class Assessment</i>	75%	1,2,3,4

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. No attempt to engage in class discussion.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Minimal engagement in discussion and debate.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Some input into discussion with evidence of independent thought.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Contribute significantly to class debate with independent thoughts.
- 70%-100%:** All the above to an excellent level. Show initiative, individual thought and enthusiasm in self-directed study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Group Work	10	
Independent Learning	89	

Supplementary Material

- Roy, J. *_An Introduction to Pharmaceutical Sciences_*. New York: Biohealthcare Publishing, 2011.

- Bennett, B. and G. Cole. _Pharmaceutical Production An Engineering Guide_. Warwickshire: IChemE, 2003.
- Sharpe, J.P. _Good Manufacturing Practice: Philosophy and Applications_. Illinois: Interpharm Press, 1991.

Requested Resources

- Lecture Room: Loose Seated